THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 12: **Beyond Earth** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If sunlight takes 8 minutes to reach the Earth, the Sun's distance in terms of light travel time is: 1

(a) 8 light-seconds **(b)** 8 light-years **(c)** 8 light-minutes **(d)** 8 light-hours

2. Mass of an object is 60 kg on Earth. On the Moon (gravity one-sixth), its mass is: 1

(a) 10 kg **(b)** 360 kg **(c)** 6 kg **(d)** 60 kg

3. Which statement correctly explains the seasons? 1

(a) Sun moves
(b) Earth's distance changes by thousands km
(c) Moon moves
(d) Earth's axis is tilted as it goes around the Sun

4. A student draws the Moon at the same time each night and sees the bright part grows. The Moon is: 1

(a) new **(b)** missing **(c)** waning **(d)** waxing

5. The weight of the same object on the Moon compared with Earth is: 1

(a) equal **(b)** one sixth **(c)** six times **(d)** zero

6. Which observation proves that the Earth rotates? 1

(a) the Moon is round
(b) stars appear to move across the night sky
(c) the Sun is hot
(d) planets are bright

7. If the new moon occurs on 1 March, the next new moon will be about: 1

(a) 8 March **(b)** 1 April + 5 days **(c)** 15 March **(d)** 30 March

8. If a satellite takes 90 minutes for one orbit, the number of orbits in 24 hours is: 1

(a) 8 **(b)** 90 **(c)** 16 **(d)** 24

9. How would the length of day change if the Earth rotated twice as fast? 1

(a) halve **(b)** stay the same **(c)** vanish **(d)** double

10. On a full-moon night the Moon rises roughly at: 1

(a) sunset **(b)** sunrise **(c)** noon **(d)** midnight

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. On a map of the solar system, 1 cm shows 150 million km. If the Sun-Earth distance is 150 million km, how far away is Mars at about 228 million km? 2

12. Why do planets appear to move against the background of stars? 2

13. Why is a launch near the equator more efficient for satellites? 2

14. A satellite goes round the Earth once in 100 minutes. How many orbits are completed in one day? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A day on planet X is 10 hours and its year is 5000 hours. How many days are there in its year? 3

16. Explain how the Sun's gravity keeps planets in orbit and what would happen if it vanished. 3

17. Explain with a diagram in words how the phases occur for the first quarter Moon. 3

OR

Moon takes about 27 days to orbit the Earth. How many orbits in 1 year (365 days)? 3

18. Why do some comets appear only once in many years? 3

OR

Why are craters on the Moon preserved for long while Earth's are not? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Design a model of the solar system for your school science fair describing the materials, scales and observations. 5

OR

Explain, using the Earth-Moon-Sun positions, why the same Moon phase can be seen all over the world on the same day but at different times. 5

20. A student records the Moon's shape for 30 days and finds it was a thin crescent on day 3, half on day 8, full on day 15, half on day 22 and none on day 29. Analyse the pattern. 5

OR

Discuss the technological spin-offs of space research for daily life and health. 5

END OF QUESTION PAPER — Total: 40 Marks